

In the image below, we have a chain of three blocks. Each block has a hash, and the hash of the previous block. Block 3 points to block 2 and block 2 in turn points to block 1.

The first block is called the Genesis block as it does not point to any other block being the first one.

Now, let's say someone tampered with the second block. This would cause the hash of the second block to change as well. In turn, this would make block 3 and all the following blocks invalid, because they no longer store a valid hash of the previous block. So making some change in a single block will make all following blocks invalid.

But using hashes is not enough to prevent tampering. Computers these days are very fast and can calculate hundreds and thousands of hashes per second. You can conclusively fiddle with the block and recalculate all the hashes of the remaining blocks to make your blockchain valid again.

To mitigate this, blockchain uses a mechanism called Proof-of-Work, which slows down the creation of new blocks. In case of bitcoin, it takes about 10 minutes to calculate the required proof-of-work and add a new block to the chain.

This mechanism makes it difficult to tamper with the blocks because if you tamper with one block, you need to recalculate the proof of work for all the following blocks.

So, we can say that the security of blockchain comes from its creative use of hashing and the proof of work mechanism.

There is one more way that blockchain secures itself and that is by being distributed.



Blockchain has inspired many new decentralised ledger systems